

Project #05 – Cisco Packet Tracer

Project #05 – Static Routing

1. Project Overview

Static Routing is a routing method in which network routes are manually configured by a network administrator. In this project, a simulated network environment was created using Cisco Packet Tracer. The network consists of two separate Local Area Networks (LANs) connected through two Cisco routers.

The first LAN represents Office A using the 192.168.10.0/24 network, while the second LAN represents Office B using the 192.168.20.0/24 network. The two LANs are connected through a dedicated point-to-point network using the 10.0.0.0/30 address range.

Because each router initially only knows about its directly connected networks, static routes must be configured manually so that both routers know how to reach the remote LAN. The main objective of this project is to understand how routers forward packets between different networks using manually configured static routes.

2. Project Objective

The objective of this project are:

- To understand the basic concept of IP routing.
- To configure IPv4 addresses on Cisco router interfaces.
- To configure communication between two routers.
- To configure static routes on Cisco routers.
- To understand directly connected and static routes.
- To analyze routing tables using Cisco IOS commands.
- To understand the concept of next-hop routing.
- To verify connectivity between different networks.
- To practice basic network troubleshooting.
- To understand packet flow between two different LAN networks.

3. Tools and Device

3.1 Software

The network simulation was developed using: **Cisco Packet Tracer**

Cisco Packet Tracer was used to simulate the network topology, configure Cisco devices, test connectivity, and analyze routing behavior.

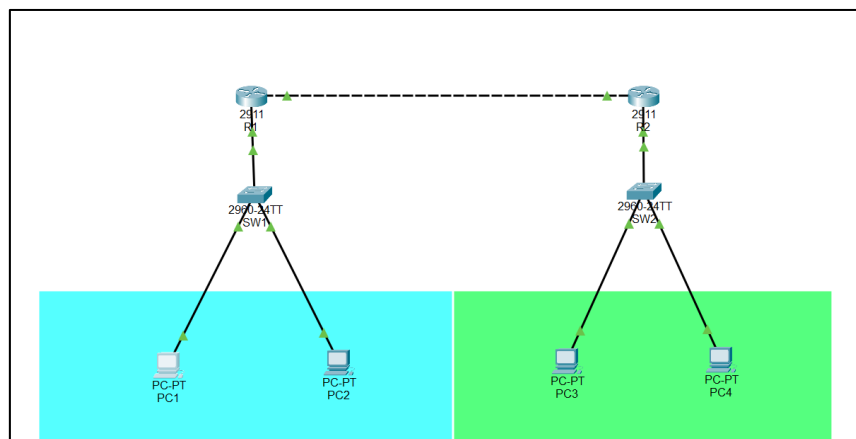
3.2 Network Devices

The following devices were used:

Device	Quantity
Cisco 2911 Router	2
Cisco 2960 Switch	2
PC	4

4. Network Topology

The network consists of two LAN environments connected through two Cisco routers. The topology can be represented as follows:



5. Network Design

The network was divided into three IPv4 networks.

Office A

Network: 192.168.10.0/24

Gateway: 192.168.10.1

Router-to-Router Link

Network: 10.0.0.0/30

R1: 10.0.0.1

R2: 10.0.0.2

Office B

Network: 192.168.20.0/24

Gateway: 192.168.20.1

The /30 subnet was selected for the router-to-router connection because a point-to-point connection only requires two usable IPv4 addresses.

6. IP Addressing Table

Device	Interface	Network	IP Address	Subnet Mask	Default Gateway
R1	G0/0	Office A	192.168.10.1	255.255.255.0	-
R1	G0/1	Router Link	10.0.0.1	255.255.255.252	-
R2	G0/1	Router Link	10.0.0.2	255.255.255.252	-
R2	G0/0	Office B	192.168.20.1	255.255.255.0	-
PC-01	Fa0	Office A	192.168.10.10	255.255.255.0	192.168.10.1
PC-02	Fa0	Office A	192.168.10.11	255.255.255.0	192.168.10.1
PC-03	Fa0	Office B	192.168.20.10	255.255.255.0	192.168.20.1
PC-04	Fa0	Office B	192.168.20.11	255.255.255.0	192.168.20.1

7. Switch Port Assignment

SW1 - Office A

Interface	Connected Device	Network
Fa0/1	PC-01	192.168.10.0/24
Fa0/2	PC-02	192.168.10.0/24
Fa0/24	R1 G0/0	192.168.10.0/24

SW2 - Office B

Interface	Connected Device	Network
Fa0/1	PC-03	192.168.20.0/24
Fa0/2	PC-04	192.168.20.0/24
Fa0/24	R2 G0/0	192.168.20.0/24

8. R1 Configuration

R1 functions as the gateway for Office A and provides connectivity toward R2.

8.1 Configure Office A Interface

```
enable
configure terminal
interface gigabitEthernet 0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
exit
```

The interface G0/0 is assigned the IP address 192.168.10.1 and functions as the default gateway for devices in Office A.

8.2 Configure Router-to-Router Interface

```
interface gigabitEthernet 0/1
ip address 10.0.0.1 255.255.255.252
no shutdown
exit
```

The interface G0/1 connects R1 to R2 through the 10.0.0.0/30 network.

9. R2 Configuration

R2 functions as the gateway for Office B and provides connectivity toward R1.

9.1 Configure Office A Interface

```
enable
configure terminal
interface gigabitEthernet 0/0
ip address 192.168.20.1 255.255.255.0
no shutdown
exit
```

The interface G0/0 functions as the default gateway for devices in Office B.

9.2 Configure Router-to-Router Interface

```
interface gigabitEthernet 0/1
ip address 10.0.0.2 255.255.255.252
no shutdown
exit
```

The interface G0/1 connects R2 to R1.

10. Static Routing Configuration

After configuring the interfaces, each router knows its directly connected networks. However, R1 does not automatically know how to reach the 192.168.20.0/24 network, and R2 does not automatically know how to reach the 192.168.10.0/24 network. Therefore, static routes were manually configured.

10.1 Static Route on R1

The following command was configured on R1:

```
ip route 192.168.20.0 255.255.255.0 10.0.0.2
```

This configuration tells R1: To reach the 192.168.20.0/24 network, forward packets to the next-hop address 10.0.0.2.

10.2 Static Route on R2

The following command was configured on R2:

```
ip route 192.168.10.0 255.255.255.0 10.0.0.1
```

This configuration tells R2: To reach the 192.168.10.0/24 network, forward packets to the next-hop address 10.0.0.1.

11. Routing Table Analysis

The routing table was verified using: show ip route

11.1 R1 Routing Table

R1 contains the following relevant routes:

```
C 10.0.0.0/30
C 192.168.10.0/24
S 192.168.20.0/24 via 10.0.0.2
```

11.2 R2 Routing Table

R2 contains:

```
C 10.0.0.0/30
C 192.168.20.0/24
S 192.168.10.0/24 via 10.0.0.1
```

12. Interface Verification

The status of the router interfaces was verified using:

show ip interface brief

This command provides information about:

- Interface name
- IP address
- Interface status
- Protocol status

13. Connectivity Testing

Connectivity testing was performed to verify that the static routes were functioning correctly.

From PC-01:

```
ping 192.168.20.10
```

The ping test successfully reached PC-03 through the two routers.

14. Reverse Connectivity Test

To verify bidirectional communication, a reverse connectivity test was performed from PC-03 to PC-01.

Command:

```
ping 192.168.10.10
```

The test verifies that R2 also has a valid route toward the Office A network.

15. Traceroute Test

Traceroute was used to analyze the path taken by packets between the two networks.

Command:

```
tracert 192.168.20.10
```

The test provides information about the intermediate network devices traversed before reaching the destination.

16. Troubleshooting Approach

A structured troubleshooting approach was applied during the implementation of the network.

The following commands were used during verification:

Interface Status

```
show ip interface brief
```

Routing Table

```
show ip route
```

Connectivity Test

```
ping <destination-ip>
```

Path Analysis

```
tracert <destination-ip>
```

This approach helps identify whether a connectivity problem originates from the physical connection, interface configuration, IP addressing, or routing configuration.

17. Static Routing Concept

The fundamental concept demonstrated in this project is that routers need a route to the destination network before they can forward packets correctly.

For example, when PC-01 sends a packet to PC-03:

Source:
192.168.10.10
Destination:
192.168.20.10

PC-01 sends the packet to its default gateway:

192.168.10.1

The packet then reaches R1. R1 checks its routing table and finds:

192.168.20.0/24
via 10.0.0.2

R1 forwards the packet to R2. R2 then forwards the packet toward:

192.168.20.10

This demonstrates the basic process of packet forwarding between different networks.

18. Testing Result

Test	Source	Destination	Result
Gateway Ping	PC-01	192.168.10.1	Successful
Router Link	R1	10.0.0.2	Successful
Inter-Network Ping	PC-01	192.168.20.10	Successful
Reverse Ping	PC-03	192.168.10.10	Successful
Traceroute	PC-01	192.168.20.10	Successful

The successful connectivity tests indicate that the configured static routes allow communication between the two LAN networks.

19. Skills Developed

Through this project, the following technical skills were practiced:

Networking Fundamentals

- IPv4 addressing
- Subnetting
- Default gateway
- Network segmentation
- Point-to-point networking

Cisco Configuration

- Cisco IOS CLI
- Router interface configuration
- Interface activation
- Static route configuration

Routing

- Directly connected routes
- Static routes
- Next-hop addresses
- Routing table analysis

Troubleshooting

- Interface verification
- Routing table inspection
- Ping testing
- Traceroute analysis
- End-to-end connectivity testing

20. Conclusion

This project successfully demonstrated the implementation of static routing using Cisco Packet Tracer.

Two separate LAN networks were connected through two Cisco 2911 routers. Each router was configured with its own LAN interface and a point-to-point connection to the other router. Static routes were then manually configured so that both routers could identify and reach the remote network. Connectivity testing using ping and traceroute demonstrated successful communication between devices located on different networks.

This project provided practical experience in understanding routing tables, next-hop addresses, static route configuration, and basic network troubleshooting. The knowledge gained from this project provides a foundation for studying more advanced routing technologies, particularly **dynamic routing protocols such as OSPF**.